



USER GUIDE

MX Shock Absorbing Mounting Rack

Revision B
May 2025

Legal information

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Release Note

This is the May 2025 release, Revision B of the *Trimble MX Shock Absorbing Mounting Rack User Guide*, P/N 37000001. It applies to:

- Trimble MX Shock Absorbing Mounting Rack.

The 37000001 original document is written in English. All documents in other languages are translations from the original English document.

Compliance Information

Europe

EC Compliance

Trimble declares that the MX Shock Absorbing Mounting Rack and associated accessories comply with the applicable directives, standards, and regulations.

WEEE

Notice to our European Customers

For product recycling instructions and more information, please go to trimble.com/en/our-commitment/responsible-business/product-compliance/environmental-compliance.

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Safety Instructions

General



The Trimble MX Shock Absorbing Mounting Rack is designed only for industrial usage. Any other usage can be dangerous and is strictly forbidden. The system must be used only by well-trained persons.



Any unauthorized changes to the product can cause personal injury or damage to the system and renders all guarantees null and void.



Any final installation of the MX Shock Absorbing Mounting Rack is under control and responsibility of the owner, or the person authorized by the owner.



It is the customer's responsibility to ensure that there are adequate mounting facilities and the component layout has been planned carefully. If necessary, the customer must obtain the approval of the local authorities before installing the MX Shock Absorbing Mounting Rack.

Mechanics



All mechanical parts must be checked before and after each mission (e.g., screws, damage on mechanical parts, etc.).



Never modify any part of the mounting rack!



If you need to change standard parts in the MX Shock Absorbing Mounting Rack (screws, washers, etc.), always use the same type of material. If you have any doubts about the parts that are needed, please contact the Trimble Support.

Usage, Installation, and Transportation



Make sure the people who are involved in installing or uninstalling the MX Shock Absorbing Mounting Rack are familiar with the "installation" chapter of this manual and have received prior training before handling the system.



All connections/screws must be tightened by complying with the torque values recommended for the used screw and mounting material!



Owing to its weight and particular handling required, the MX Shock Absorbing Mounting Rack should be installed by two operators.



The maximum speed of a vehicle fitted with the MX Shock Absorbing Mounting Rack should not exceed 110 km/h (68 mph).



The driver should remember that the MX Shock Absorbing Mounting Rack adds additional height to the vehicle and must be aware of the total height of the vehicle!



Avoid working overhead. Use tools to get a better position for installation.



Switch off the vehicle engine during MX Shock Absorbing Mounting Rack installation.



In order to avoid the danger of suffocation, keep plastic foils, bags and bubble wraps away from babies and children. Do not use them in or near children's beds, pushchairs or playpens.

Packaging materials are not a toy, so please tie them up before disposing of them.



Keep product surfaces clean and dry, cleaning the mounting rack after each mission.

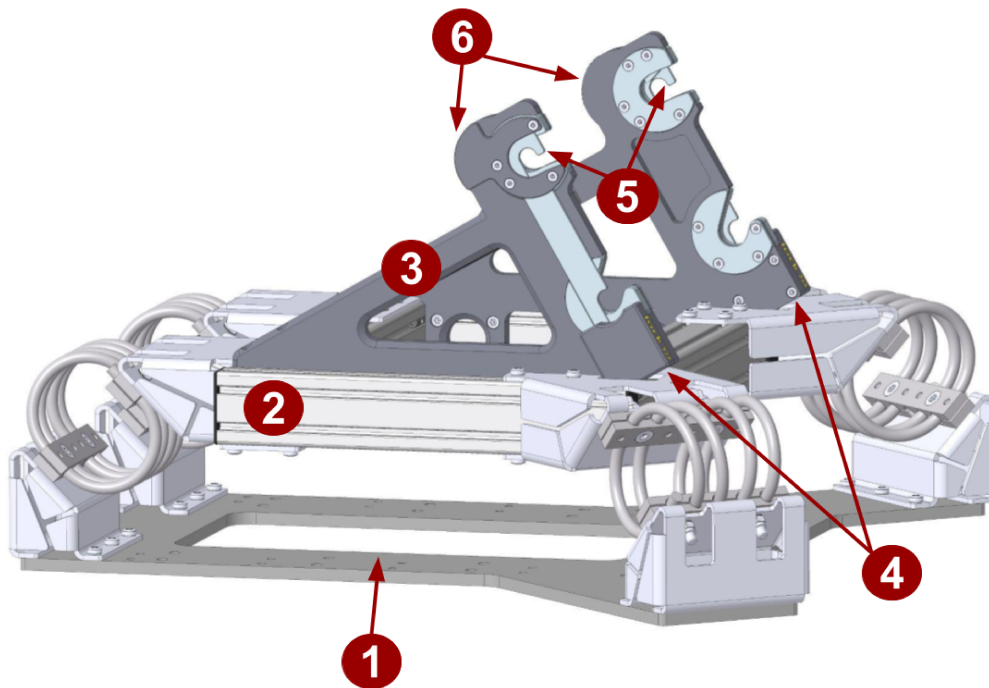
Overview and Installation

The Trimble MX Shock Absorbing Mounting Rack is a mounting facility for the MX sensor units of Mobile Mapping Systems and is designed to be mounted on vehicles mostly driving in harsher environments than a standard road.

The base plate of the MX Shock Absorbing Mounting Rack includes two types of screw holes with two different screw hole diameters to mount onto the prepared mounting location.

The MX Shock Absorbing Mounting Rack consists of the following components:

- 1 Base plate:** Used to fix the MX Shock Absorbing Mounting Rack on a vehicle. There are two different screw holes diameters for mounting.
- 2 Mounting rack mainframe:** Is the main structure of the MX Shock Absorbing Mounting Rack including shock absorbers.
- 3 Mounting facilities** for the MX sensor units.
- 4 Safety locks:** Used to firmly secure the MX sensor unit.
- 5 Lock bars:** Used in conjunction with the safety locks.
- 6 Fast locks:** Latches that are automatically locked in once the MX sensor unit has been properly inserted.



Basic requirement for the mounting location

The mounting location on the vehicle needs to be specified in a way to ensure that the weight of the MX Shock Absorbing Mounting Rack (28 kg) and the MX sensor unit (up to 37 kg) keeps safe under all driving conditions (including forces of shocks and vibrations)

The mounting platform must be leveled in a horizontal position.

There are two ways mounting the MX Shock Absorbing Mounting Rack on the mounting platform or aluminum profiles:

- Using M6 or 1/4 inch screws for the 20 holes with a diameter of 6.6 mm (blue marked).

The arrangement of the 6.6 mm holes are designed to fit to typical aluminum profiles (60 mm × 120 mm) with parallel T-slots.

- Using M8 or 5/16 inch screws for the 16 holes with a diameter of 9.0 mm (red marked).

The arrangement of the 9.0 mm holes are designed to fit to typical aluminum profiles (80 mm × 160 mm) with parallel T-slots.

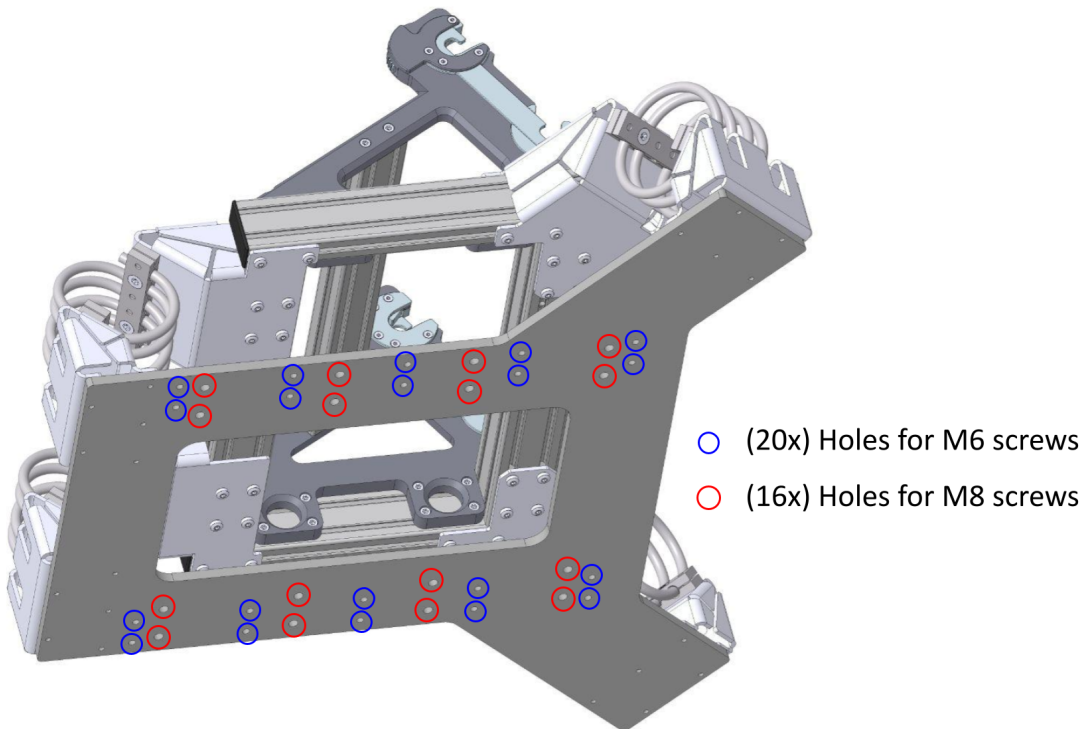


All holes for one kind of screw must be used.



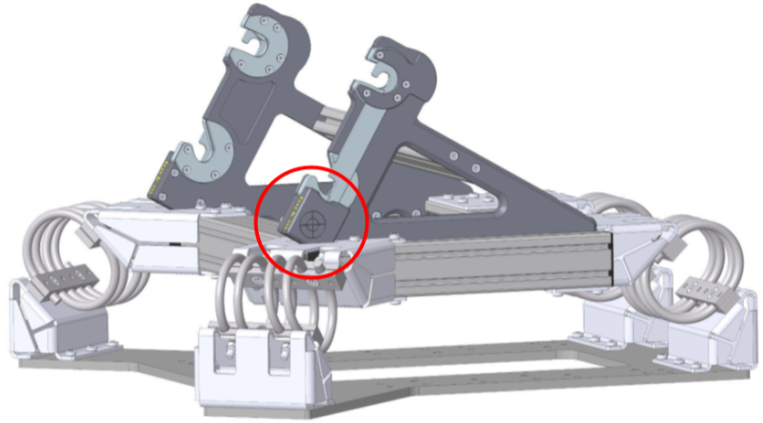
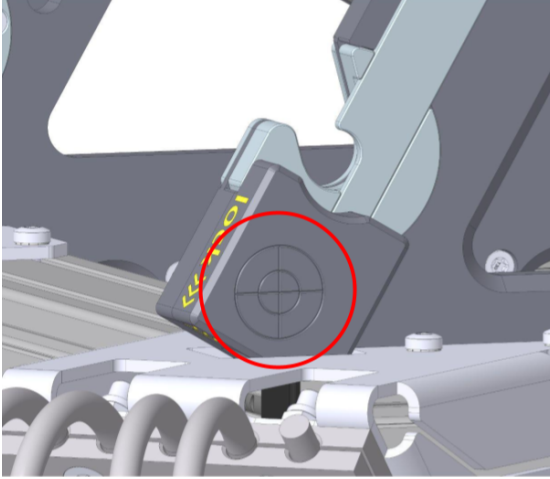
The strength for the screws must be minimum A2-70 for stainless steel or class 8.8 for steel.

The following drawing shows the positions of the holes. The precise values for positions and sizes are available in [Dimensions, page 9](#).



External Reference Point

The MX Shock Absorbing Mounting Rack also carries the external reference point of the system for lever-arm measurements. For more information, refer to the user guide for your Trimble MX system (Operation chapter: External reference point).



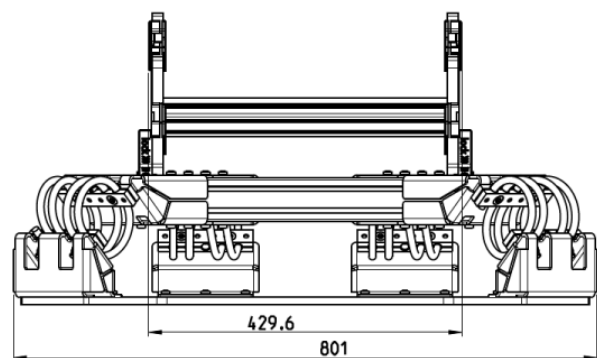
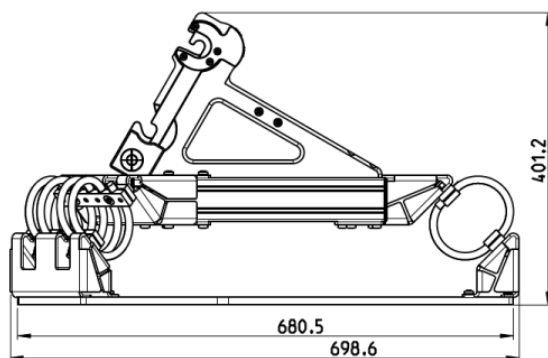
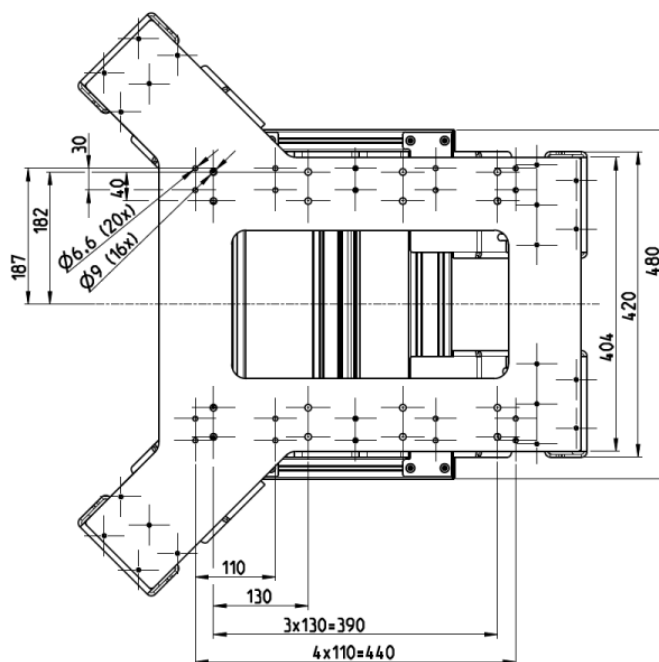
Technical Specifications

Weight

The weight of the MX Shock Absorbing Mounting Rack is 28 kg.

Dimensions

All dimensions are shown in mm.



Support

General

If you need support for the Trimble MX System or Trimble Mobile Mapping software, please contact our Customer Support team via email:

Trimble Support Email **mx_support@trimble.com**

Or call:

Support Americas: +1-289-695-4416

Support APAC: +86-105-603-4179

Support Europe & Rest of World: +49-7351-474-0237

Please report your support case as precisely as possible. Please provide:

- a short description of the problem.
- the workflow you used and how to reproduce the problem.
- If occurring during a mission:
 - the location of the mission.
 - environmental conditions.

Also send the following information about your Trimble MX System:

- System log file.
- Serial number.
- Total operating hours of your Trimble MX System since purchase.
- Photos or videos may also be useful to describe the problem.

To register your product or check the status of your system maintenance, please register your system and software at <https://mytrimbleprotected.com/>.

My Trimble Protected enables Trimble users, channel partners, and end customers to streamline business processes for customer engagement and service. As a centralized instrument panel, My Trimble Protected manages registrations, serial number lookups, product catalogs, reports, settings, and locator processes. It allows users to configure, monitor, and optimize each aspect of the process.

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Trimble Land Mobile Download Center

System software updates, FAQs, and documentation are available in the Trimble Land Mobile Download Center at geospatial.trimble.com.